INTRODUCTION TO INFORMATION GEOMETRY

Cyclic Motion and Invasion

Week 3 · Lecture 2

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One family, three kinds of motion

For

$$M_s = \begin{pmatrix} 0 & 1 & -s \\ -s & 0 & 1 \\ 1 & -s & 0 \end{pmatrix}, \qquad s > 0,$$

the equal mixture $\mathbf{p}^* = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ is an equilibrium. As s varies, trajectories may approach the center, remain on closed curves, or move toward the boundary.

A scalar quantity for cyclic motion

For an interior population set

$$P = p_A p_B p_C, \qquad Q = p_A p_B + p_B p_C + p_C p_A.$$

The product satisfies

$$0 < P \leq \frac{1}{27},$$

with equality only at the equal mixture. The motion of P detects movement toward or away from the center.

Differentiating the product

Take $P = p_A p_B p_C > 0$. Differentiate the logarithm:

$$\frac{d}{dt} \ln P = \sum_i \frac{p_i'}{p_i} = \sum_i (f_i - \bar{f}).$$

For the cyclic family, $\sum_i f_i = 1 - s$ and $\bar{f} = (1 - s)Q$. Find $\frac{d}{dt} \ln P$.

Differentiating the product

Take $P = p_A p_B p_C > 0$. Differentiate the logarithm:

$$\frac{d}{dt} \ln P = \sum_i \frac{p_i'}{p_i} = \sum_i (f_i - \bar{f}).$$

Using $\sum_i f_i = 1 - s$ and $\bar{f} = (1 - s)Q$,

$$\frac{d}{dt} \ln P = (1 - s)(1 - 3Q).$$

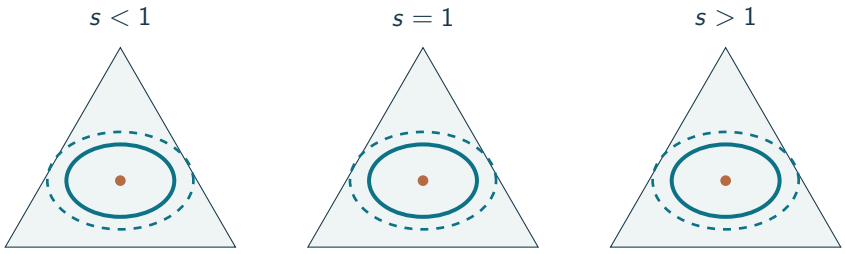
The second factor is nonnegative

Because $p_A + p_B + p_C = 1$,

$$1 - 3Q = \frac{1}{2}\Big[(p_A - p_B)^2 + (p_B - p_C)^2 + (p_C - p_A)^2\Big].$$

Thus $1 - 3Q > 0$ away from the equal mixture. The sign of $\frac{d}{dt} \ln P$ is controlled by $1 - s$.

Three regimes around the center



The ellipses schematically show nested product levels.

What the product proves

- $s < 1$: P increases toward its maximum $1/27$. Interior motion approaches the center.
- $s = 1$: P is conserved. Noncentral interior solutions are periodic.
- $s > 1$: P decreases toward zero. Interior motion approaches the boundary as a set.

Linearizing the cyclic system

Use $x = p_A$, $y = p_B$, and $p_C = 1 - x - y$. At the center, the Jacobian is

$$J_s = \frac{1}{3}\begin{pmatrix} s & s+1 \\ -(s+1) & -1 \end{pmatrix}.$$

Its eigenvalues are

$$\lambda_\pm = \frac{s-1}{6} \pm i\frac{\sqrt{3}(s+1)}{6}.$$

A damped cycle

Take $s = 0.7$. From

$$\lambda_{\pm} = \frac{s-1}{6} \pm i\frac{\sqrt{3}(s+1)}{6},$$

find the decay rate ρ, angular frequency ω, and approximate small-oscillation period $T = 2\pi/\omega$.

A damped cycle

Take $s = 0.7$. The eigenvalues have

$$\rho = -0.05, \qquad \omega \approx 0.491.$$

The small-oscillation period is

$$T = \frac{2\pi}{\omega} \approx 12.80.$$

The linear envelope decays like $e^{-0.05t}$.

Stability within a face

A population on the AB edge can attract all other states of that edge. To assess stability in the full triangle, we must also ask whether a small number of C individuals can grow. At a face equilibrium q, the initial growth of a missing type is

$$p_C' \approx p_C\big(f_C(q) - \bar{f}(q)\big).$$

An invader changes the stability question

Let

$$M = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 0 \\ 3 & 3 & 0 \end{pmatrix}, \qquad q = (\tfrac{1}{2}, \tfrac{1}{2}, 0).$$

At q, find $\bar{f}$ and C's invasion rate $f_C(q) - \bar{f}(q)$.

An invader changes the stability question

Let

$$M = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 0 \\ 3 & 3 & 0 \end{pmatrix}, \qquad q = (\tfrac{1}{2}, \tfrac{1}{2}, 0).$$

The fitnesses at q are $(1, 1, 3)$, hence $\bar{f} = 1$.

$$f_C(q) - \bar{f}(q) = 3 - 1 = 2 > 0.$$

An invariant edge may be unstable in the triangle

At $q = (\frac{1}{2}, \frac{1}{2}, 0)$, the AB-edge equation is

$$x' = x(1-x)(2-4x).$$

Its derivative at $x = \frac{1}{2}$ is -1: attraction along the edge. But a small C introduction has

$p'_C \approx 2p_C$, so q repels in that transverse direction.

From populations to probability laws

Every normalized vector is a categorical distribution.

$$\mathbf{p}(t) = (p_A(t), p_B(t), p_C(t)).$$

The simplex is simultaneously a state space for a dynamical system and a space of possible sampling probabilities. Next week, we study that probability interpretation directly.